

Article

An Integrable Nonlinear Schrödinger System with Symmetric Matrix Potentials and Its Binary Darboux Transformations

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Abstract

This study aims to propose a class of binary Darboux transformations for an integrable nonlinear Schrödinger system with two symmetric matrix potentials. The associated Lax pairs of AKNS type ensure the existence of these binary Darboux transformations, whose explicit single-step applications yield soliton solutions. Notably, the M -matrix appearing in the formulation of the binary Darboux transformations must be constructed within an extended framework, allowing for cases where eigenvalues coincide with adjoint eigenvalues, thereby encompassing generalized Darboux transformations as well.

Keywords: Lax pair; zero-curvature equation; binary Darboux transformation

MSC: 37K10; 35Q55; 37K35

1. Introduction

Integrable models play a fundamental role in the study of important coherent structures, such as solitons, breathers, lump waves, and rogue waves. These systems are often constructed from Lax pairs consisting of spatial and temporal matrix eigenvalue problems [1]. Based on such Lax pairs, a variety of analytical techniques have been systematically developed for solving integrable models, including the inverse scattering transform, Darboux transformations, and the Riemann–Hilbert approach (see, e.g., [2–5]).

A Lax pair defines two linear eigenvalue problems:

$$\phi_x = U\phi = U(u, \lambda)\phi, \quad \phi_t = V\phi = V(u, \lambda)\phi, \quad (1)$$

where U and V are square matrices, $u = u(x, t)$ is the potential, and λ is the spectral parameter. The Lax pair determines an integrable model [6], through the compatibility condition $\phi_{xt} = \phi_{tx}$, which is equivalent to the zero-curvature equation:

$$U_t - V_x + [U, V] = 0. \quad (2)$$

The corresponding adjoint matrix eigenvalue problems are defined by

$$\psi_x = -\psi U, \quad \psi_t = -\psi V, \quad (3)$$

whose compatibility condition $\psi_{xt} = \psi_{tx}$ leads to the same zero-curvature equation as in (2).



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In this study, we consider an integrable nonlinear Schrödinger (NLS) system with two symmetric matrix potentials p and q :

$$p_t = \frac{1}{2}(p_{xx} - 2pqp), \quad q_t = -\frac{1}{2}(q_{xx} - 2qpq), \quad p^T = p, \quad q^T = q. \quad (4)$$

This system is associated with the following Lax pair, which belongs to the symplectic Lie algebra $\mathfrak{sp}(2m)$:

$$\begin{cases} U = \begin{bmatrix} \lambda I_m & p \\ q & -\lambda I_m \end{bmatrix}, \\ V = \lambda^2 \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix} + \lambda \begin{bmatrix} 0 & p \\ q & 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} pq & -p_x \\ q_x & -qp \end{bmatrix}, \end{cases} \quad (5)$$

where I_m is the $m \times m$ identity matrix, and p and q are $m \times m$ symmetric matrices.

A binary Darboux transformation (DT) is implemented via a gauge transformation (see, e.g., [7]):

$$\phi' = S^+ \phi, \quad \psi' = \psi S^-, \quad S^+ = S^+(u, \lambda), \quad S^- = S^-(u, \lambda), \quad (6)$$

where S^+ and S^- are mutual inverses and can be derived from an M -matrix formulation (see, e.g., [7] or Section 3 for a general framework). The key transformation conditions are:

$$\begin{cases} S_x^+(S^+)^{-1} + S^+ U (S^+)^{-1} = U(u', \lambda), \\ S_t^+(S^+)^{-1} + S^+ V (S^+)^{-1} = V(u', \lambda), \end{cases} \quad (7)$$

where

$$u' = f(u). \quad (8)$$

Consequently, ϕ' and ψ' form a new pair of eigenfunction and adjoint eigenfunction, satisfying the eigenvalue problems:

$$\phi'_x = U' \phi', \quad \phi'_t = V' \phi', \quad (9)$$

and

$$\psi'_x = -\psi' U', \quad \psi'_t = -\psi' V', \quad (10)$$

where U' and V' retain the same structure as U and V , namely,

$$U' = U(u', \lambda), \quad V' = V(u', \lambda). \quad (11)$$

These transformed eigenfunctions ensure that the new Lax pair (U', V') still satisfies the original zero-curvature equation, with u being replaced by u' . Hence, the transformed potential u' provides another solution to the integrable model associated with the given Lax pair (U, V) .

In soliton theory, binary DTs have been widely applied to integrable models, including the NLS system, the modified Korteweg–de Vries system and the modified Kadomtsev–Petviashvili equation (see, e.g., [8–11]), particularly for constructing N -soliton solutions, as well as breather, rogue, and lump wave solutions. In the case of Lax pairs with reductions, it has been noted recently (see, e.g., [8]) that orthogonality conditions between eigenfunctions and adjoint eigenfunctions must be imposed, especially when eigenvalues coincide with adjoint eigenvalues.

In this paper, we propose a class of binary DTs for the matrix integrable NLS system with two symmetric potentials introduced above. These transformations are constructed via an M -matrix generating procedure, which requires an extended formulation of the M -matrix to accommodate the case where eigenvalues may coincide with adjoint eigenvalues. Single applications of the proposed transformations yield explicit soliton solutions. The paper concludes with a summary of the main results and concluding remarks.

2. An Integrable NLS System with Symmetric Matrix Potentials

Let m be an arbitrary natural number. We consider the following coupled system of integrable NLS equations:

$$\begin{cases} p_t = \frac{1}{2}(p_{xx} - 2pq p), \\ q_t = -\frac{1}{2}(q_{xx} - 2qp q), \end{cases} \quad (12)$$

where p and q are two $m \times m$ symmetric matrices. This model is generated from the Lax pair of matrix eigenvalue problems:

$$\phi_x = U\phi, \quad \phi_t = V\phi, \quad (13)$$

where U and V are associated with the symplectic Lie algebra $\mathfrak{sp}(2m)$ and are given by (5), namely,

$$U = U(u, \lambda) = \lambda\Lambda + P, \quad \Lambda = \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix}, \quad P = \begin{bmatrix} 0 & p \\ q & 0 \end{bmatrix}, \quad (14)$$

and

$$V = V(u, \lambda) = \lambda^2\Lambda + \lambda P - \frac{1}{2}\Lambda(P^2 - P_x), \quad (15)$$

with $u = u(p, q)$. The symmetric conditions $p^T = p$ and $q^T = q$ are equivalent to the reduction

$$\Delta P \Delta^{-1} = -P^T, \quad \Delta = \begin{bmatrix} 0 & I_m \\ -I_m & 0 \end{bmatrix}, \quad (16)$$

which ensures that P , and hence U and V , lie in the symplectic Lie algebra $\mathfrak{sp}(2m)$.

The coupled integrable NLS system (12) possesses an infinite number of commuting flows:

$$u_t = X^{[k]} = X^{[k]}(u) = \begin{bmatrix} 2b^{[k+1]} \\ -2c^{[k+1]} \end{bmatrix}, \quad k \geq 0, \quad (17)$$

which are generated from the Lax pair $(U, V^{[k]})$, where

$$V^{[k]} = \sum_{l=0}^k \lambda^{k-l} W^{[l]}, \quad W^{[l]} = \begin{bmatrix} a^{[l]} & b^{[l]} \\ c^{[l]} & d^{[l]} \end{bmatrix}. \quad (18)$$

Here, the Laurent series

$$W = \sum_{l \geq 0} W^{[l]} \lambda^{-l}, \quad W^{[0]} = \Lambda, \quad (19)$$

is assumed to solve the stationary zero-curvature equation $W_x = [U, W]$. To uniquely determine the hierarchy, we impose the normalization condition $W^{[l]}|_{u=0} = 0$, $l \geq 1$ (see, e.g., [12–14]). The member corresponding to $k = 2$ is precisely the matrix integrable NLS model (12), since

$$V = \lambda^2 W^{[0]} + \lambda W^{[1]} + W^{[2]}. \quad (20)$$

The commutativity of these flows,

$$[[X^{[k]}, X^{[l]}]] := X^{[k]'}(u)[X^{[l]}] - X^{[l]'}(u)[X^{[k]}] = 0, \quad k, l \geq 0, \quad (21)$$

can be verified using the Lax operator algebra (see, e.g., [12]).

The integrable NLS model (12) encompasses typical examples of coupled integrable systems and further extends the class of multi-component integrable NLS-type models (see, e.g., [15–18]). Through appropriate reductions, one can derive scalar or matrix integrable equations. For example, by taking

$$p(t, x) = \frac{\sqrt{2}}{2}v(\frac{1}{2}it, ix), \quad q(t, x) = \pm \frac{\sqrt{2}}{2}v^\dagger(\frac{1}{2}it, ix), \quad (22)$$

where $\dagger$ denotes the Hermitian conjugate, the model (12) reduces to the matrix focusing or defocusing NLS equation:

$$iv_t + v_{xx} \pm vv^\dagger v = 0, \quad (23)$$

where v is a square matrix. A linear combination of this equation with the first linear member of the hierarchy (corresponding to $k = 0$) provides a mathematical framework for describing soliton formation and evolution in fiber lasers (see, e.g., [19]). Furthermore, when v is scalar, the matrix equation further reduces to the standard scalar focusing and defocusing NLS equation:

$$iv_t + v_{xx} \pm |v|^2 v = 0, \quad (24)$$

which has various applications in nonlinear optics [20].

3. Binary Darboux Transformations

For the matrix integrable NLS system (12), associated with the Lax pair (13), we aim to construct a class of binary DTs. This is achieved by employing pairs of eigenfunctions and adjoint eigenfunctions, together with an M -matrix formulation, to systematically generate the binary DTs.

3.1. The Unreduced Case

Let N be an arbitrary natural number and let u be a given solution of the integrable NLS model (12). For the Lax pair (13), with U and V defined by (14) and (15), respectively, we construct a class of binary DTs based on N pairs of eigenfunctions and adjoint eigenfunctions, in the case of unreduced matrix potentials p and q .

Assume that, for given eigenvalues λ_i and adjoint eigenvalues $\hat{\lambda}_i$ ($1 \leq i \leq N$), the corresponding eigenfunctions v_i and adjoint eigenfunctions $\hat{v}_i$ satisfy the matrix eigenvalue problems:

$$v_{i,x} = U(u, \lambda_i)v_i, \quad v_{i,t} = V(u, \lambda_i)v_i, \quad 1 \leq i \leq N, \quad (25)$$

$$\hat{v}_{i,x} = -\hat{v}_i U(u, \hat{\lambda}_i), \quad \hat{v}_{i,t} = -\hat{v}_i V(u, \hat{\lambda}_i), \quad 1 \leq i \leq N, \quad (26)$$

where U and V are defined by (14) and (15), respectively. These eigenfunctions and adjoint eigenfunctions serve as the fundamental building blocks for constructing N -fold binary DTs. This framework enables the generation of new solutions to the coupled integrable NLS system (12) through a superposition principle.

A crucial step in constructing the N -fold binary DTs is the introduction of an $N \times N$ matrix $M = (m_{ij})$, whose entries encode the interactions between the eigenfunctions and adjoint eigenfunctions [10]. Specifically, we define

$$M = (m_{ij})_{N \times N}, m_{ij} = \begin{cases} \frac{\hat{v}_i v_j}{\lambda_j - \hat{\lambda}_i}, & \text{if } \lambda_j \neq \hat{\lambda}_i, \\ m_{ij}^c(x, t), & \text{if } \lambda_j = \hat{\lambda}_i, \end{cases} \text{ where } 1 \leq i, j \leq N, \quad (27)$$

where v_i and $\hat{v}_i$ are the eigenfunctions and adjoint eigenfunctions introduced in (25) and (26). The entries $m_{ij}^c(x, t)$ are placeholders for the coinciding-eigenvalue case $\lambda_j = \hat{\lambda}_i$, defined by

$$m_{ij,x}^c = \hat{v}_i \Theta_{ij} v_j, \text{ if } \lambda_j = \hat{\lambda}_i, \quad (28)$$

and

$$m_{ij,t}^c = \hat{v}_i \Xi_{ij} v_j, \text{ if } \lambda_j = \hat{\lambda}_i, \quad (29)$$

where $1 \leq i, j \leq N$. Here

$$\Theta_{ij} = \frac{U(u, \lambda_j) - U(u, \hat{\lambda}_i)}{\lambda_j - \hat{\lambda}_i} = \Lambda, \quad \Xi_{ij} = \frac{V(u, \lambda_j) - V(u, \hat{\lambda}_i)}{\lambda_j - \hat{\lambda}_i} = (\lambda_j + \hat{\lambda}_i)\Lambda + P. \quad (30)$$

This M -matrix plays a central role in forming the higher-rank projector that defines the binary Darboux matrices.

When the $N \times N$ matrix M defined in (27) is invertible, we can construct the N -fold binary DTs as follows. More precisely, the Darboux matrices are defined by

$$\begin{cases} S^+ = S^+(\lambda) = I_{2m} - \sum_{i,j=1}^N \frac{v_i (M^{-1})_{ij} \hat{v}_j}{\lambda - \hat{\lambda}_j}, \\ S^- = S^-(\lambda) = I_{2m} + \sum_{i,j=1}^N \frac{v_i (M^{-1})_{ij} \hat{v}_j}{\lambda - \lambda_i}, \end{cases} \quad (31)$$

together with their leading-order terms at infinity:

$$S_1^\pm = \lim_{\lambda \rightarrow \infty} [\lambda(S^\pm(\lambda) - I_{2m})]. \quad (32)$$

Here v_i and $\hat{v}_j$ are the eigenfunctions and adjoint eigenfunctions defined in (25) and (26), respectively.

Based on the binary DT theory, we need to verify the following two gauge conditions. The first condition is

$$S_x^+ = U' S^+ - S^+ U, \quad (33)$$

where

$$U' = \lambda \Lambda + P', \quad (34)$$

with

$$P' = \begin{bmatrix} 0 & p' \\ q' & 0 \end{bmatrix} = P + S_1^+ \Lambda + \Lambda S_1^-. \quad (35)$$

The second condition is

$$S_t^+ = V' S^+ - S^+ V, \quad (36)$$

where

$$V' = \lambda^2 \Lambda + \lambda P' - \frac{1}{2} \Lambda (P'^2 - P_x'). \quad (37)$$

Verification of (33) and (36) ensures that a class of binary DTs can be established for the coupled integrable NLS system (12).

We remark that if the eigenvalues satisfy the non-degeneracy condition

$$\{\lambda_i | 1 \leq i \leq N\} \cap \{\hat{\lambda}_i | 1 \leq i \leq N\} = \emptyset; \lambda_i \neq \lambda_j, \hat{\lambda}_i \neq \hat{\lambda}_j, 1 \leq i, j \leq N; \quad (38)$$

then (31) reduces to Darboux matrices in the standard N -fold DT formulation, which has been extensively analyzed in the literature (see, e.g., [3,5,21–23]). If some eigenvalues coincide with adjoint eigenvalues, then the matrix M must be chosen with care, and the resulting construction leads to more intricate solutions [10]. Such situations naturally arise in the context of spectral matrices associated with symplectic Lie algebras.

We can now readily establish the following three propositions, as has been done in many other cases (see, e.g., [8,10]).

Proposition 1. *For the Darboux matrices defined in (31), the following properties hold:*

$$S^+(\lambda_i)v_i = 0, \hat{v}_i S^-(\hat{\lambda}_i) = 0, 1 \leq i \leq N, \quad (39)$$

provided that $S^+(\lambda_i)$ and $S^-(\hat{\lambda}_i)$ are well-defined.

Employing the formulas for the two Darboux matrices S^+ and S^- in (31), one can readily derive their partial fraction decompositions, as summarized in the following proposition.

Proposition 2. *Let S^+ and S^- be the Darboux matrices defined by (31). Then their partial fraction decompositions are given by*

$$S^+ = I_{2m} - \sum_{i=1}^N \frac{v_i^M \hat{v}_i}{\lambda - \hat{\lambda}_i}, S^- = I_{2m} + \sum_{i=1}^N \frac{v_i \hat{v}_i^M}{\lambda - \lambda_i} \quad (40)$$

where the vectors v_i^M and $\hat{v}_i^M$ are defined in terms of the $N \times N$ matrix M as

$$\begin{cases} (v_1^M, v_2^M, \dots, v_N^M)M = (v_1, v_2, \dots, v_N), \\ M((\hat{v}_1^M)^T, (\hat{v}_2^M)^T, \dots, (\hat{v}_N^M)^T)^T = (\hat{v}_1^T, \hat{v}_2^T, \dots, \hat{v}_N^T)^T. \end{cases} \quad (41)$$

From the above proposition, we can further deduce the following result.

Proposition 3. *The Darboux matrices S^+ and S^- satisfy*

$$S^+(\lambda)S^-(\lambda) = I_{2m}, S_1^+ = -S_1^-, \quad (42)$$

provided the orthogonality condition

$$\hat{v}_i v_j = 0, \text{ if } \lambda_j = \hat{\lambda}_i, \text{ where } 1 \leq i, j \leq N, \quad (43)$$

is satisfied.

The condition (43) is necessary when some eigenvalues coincide with adjoint eigenvalues. In particular, the first equality in (42) ensures that $S^+(\lambda)$ and $S^-(\lambda)$ are mutual inverses.

For clarity, we define the block vectors:

$$v = (v_1, v_2, \dots, v_N), \hat{v} = (\hat{v}_1^T, \hat{v}_2^T, \dots, \hat{v}_N^T)^T, \quad (44)$$

and the diagonal matrices

$$\begin{cases} F = \text{diag}(\frac{1}{\lambda-\lambda_1}, \frac{1}{\lambda-\lambda_2}, \dots, \frac{1}{\lambda-\lambda_N}), \\ \hat{F} = \text{diag}(\frac{1}{\lambda-\hat{\lambda}_1}, \frac{1}{\lambda-\hat{\lambda}_2}, \dots, \frac{1}{\lambda-\hat{\lambda}_N}). \end{cases} \quad (45)$$

Using these notations, the Darboux matrices take the compact form:

$$S^+ = I_{2m} - vM^{-1}\hat{F}\hat{v}, \quad S^- = I_{2m} + vFM^{-1}\hat{v}. \quad (46)$$

One can directly verify the key identity

$$\hat{v}v = \hat{F}^{-1}M - MF^{-1}, \quad (47)$$

which holds entrywise. Using this, it is straightforward to check that

$$S^+S^- = I_{2m} - vM^{-1}\hat{F}\hat{v} + vFM^{-1}\hat{v} - vM^{-1}\hat{F}\hat{v}vFM^{-1}\hat{v} = I_{2m}.$$

Taking the limits at infinity gives the leading-order terms:

$$S_1^+ = - \sum_{i,j=1}^N (M^{-1})_{ij}v_i\hat{v}_j = -vM^{-1}\hat{v}, \quad S_1^- = \sum_{i,j=1}^N (M^{-1})_{ij}v_i\hat{v}_j = vM^{-1}\hat{v}, \quad (48)$$

which immediately shows that $S_1^+ = -S_1^-$.

At this stage, we are ready to present a class of general binary DTs for the coupled integrable NLS system (12), without imposing the symmetric reductions on p and q .

Theorem 1. Let v_i and $\hat{v}_i$ ($1 \leq i \leq N$) be defined by (25) and (26), respectively, and let the M -matrix be given by (27), with the entries m_{ij}^c defined via (28)–(30). Assume that the orthogonality condition (43) holds. Then the coupled integrable NLS system (12) admits the following binary DT:

$$\phi' = S^+\phi, \quad \psi' = \psi S^-, \quad P' = P + S_1^+\Lambda + \Lambda S_1^-, \quad (49)$$

where $S^\pm$ and $S_1^\pm$ are defined by (31) and (32), or equivalently by (46) and (48).

Proof. Assume that U' and V' are defined by (34) and (37), respectively, with P' given by (35). Our proof consists in verifying the two gauge conditions (33) and (36).

We first verify the spatial condition (33). From the eigenvalue problems, we have

$$v_x = \Lambda v D + P v, \quad \hat{v}_x = -(\hat{D}\hat{v}\Lambda + \hat{v}P), \quad (50)$$

where

$$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N), \quad \hat{D} = \text{diag}(\hat{\lambda}_1, \hat{\lambda}_2, \dots, \hat{\lambda}_N). \quad (51)$$

Using the definition of M in (27) together with (28) and (29), we obtain

$$M_x = \hat{v}\Theta v, \quad \Theta = (\Theta_{ij})_{N \times N} = (\Lambda)_{N \times N}, \quad (52)$$

and

$$M_t = \hat{v}\Xi v, \quad \Xi = (\Xi_{ij})_{N \times N} = ((\lambda_j + \hat{\lambda}_i)\Lambda + P)_{N \times N} = \hat{D}\Lambda + \Lambda D + (P)_{N \times N}. \quad (53)$$

Next, using the compact forms (46) and (48) for S^+ and S_1^+ , together with the derivative formulas (50) and (52), a direct computation gives

$$\begin{aligned}
S_x^+ &= (-\Lambda v D M^{-1} \hat{F} \hat{\vartheta} - P v M^{-1} \hat{F} \hat{\vartheta}) + v M^{-1} \hat{\vartheta} \Lambda v M^{-1} \hat{F} \hat{\vartheta} \\
&\quad + (v M^{-1} \hat{F} \hat{D} \hat{\vartheta} \Lambda + v M^{-1} \hat{F} \hat{\vartheta} P), \\
U' S^+ &= (\lambda \Lambda + P - v M^{-1} \hat{\vartheta} \Lambda + \Lambda v M^{-1} \hat{\vartheta}) - (\lambda \Lambda v M^{-1} \hat{F} \hat{\vartheta} + P v M^{-1} \hat{F} \hat{\vartheta} \\
&\quad - v M^{-1} \hat{\vartheta} \Lambda v M^{-1} \hat{F} \hat{\vartheta} + \Lambda v M^{-1} \hat{\vartheta} v M^{-1} \hat{F} \hat{\vartheta}), \\
S^+ U &= (\lambda \Lambda + P) - (\lambda v M^{-1} \hat{F} \hat{\vartheta} \Lambda + v M^{-1} \hat{F} \hat{\vartheta} P).
\end{aligned}$$

Combining these expressions and using the identities

$$\hat{D} = -\hat{F}^{-1} + \lambda I_N, \quad D = -F^{-1} + \lambda I_N, \quad (54)$$

we find

$$\begin{aligned}
&S_x^+ - U' S^+ + S^+ U \\
&= -\Lambda v D M^{-1} \hat{F} \hat{\vartheta} + v M^{-1} \hat{F} \hat{D} \hat{\vartheta} \Lambda + v M^{-1} \hat{\vartheta} \Lambda - \Lambda v M^{-1} \hat{\vartheta} \\
&\quad + \lambda \Lambda v M^{-1} \hat{F} \hat{\vartheta} + \Lambda v M^{-1} \hat{\vartheta} v M^{-1} \hat{F} \hat{\vartheta} - \lambda v M^{-1} \hat{F} \hat{\vartheta} \Lambda \\
&= \Lambda v F^{-1} M^{-1} \hat{F} \hat{\vartheta} - \Lambda v M^{-1} \hat{\vartheta} + \Lambda v M^{-1} \hat{\vartheta} v M^{-1} \hat{F} \hat{\vartheta} = 0,
\end{aligned}$$

where the last equality follows from (47). Thus, the spatial condition (33) is satisfied.

The temporal condition (36) can be verified in an analogous manner.

Note that we have

$$v_t = \Lambda v D^2 + P v D + W^{[2]} v, \quad \hat{v}_t = -(\hat{D}^2 \hat{\vartheta} \Lambda + \hat{D} \hat{\vartheta} P + \hat{\vartheta} W^{[2]}), \quad (55)$$

where $W^{[2]}$ is defined by

$$W^{[2]} = V - \lambda^2 \Lambda - \lambda P = -\frac{1}{2} \Lambda (P^2 - P_x). \quad (56)$$

A direct computation yields

$$\begin{aligned}
S_t^+ &= -v_t M^{-1} \hat{F} \hat{\vartheta} + v M^{-1} M_t M^{-1} \hat{F} \hat{\vartheta} - v M^{-1} \hat{F} \hat{\vartheta}_t \\
&= -(\Lambda v D^2 + P v D + W^{[2]} v) M^{-1} \hat{F} \hat{\vartheta} \\
&\quad + v M^{-1} (\hat{\vartheta} \hat{D} \Lambda v + \hat{\vartheta} \Lambda D v + \hat{\vartheta} P v) M^{-1} \hat{F} \hat{\vartheta} \\
&\quad + v M^{-1} \hat{F} (\hat{D}^2 \hat{\vartheta} \Lambda + \hat{D} \hat{\vartheta} P + \hat{\vartheta} W^{[2]}),
\end{aligned}$$

and

$$\begin{aligned}
V' S^+ - S^+ V &= (\lambda^2 \Lambda + \lambda P' + W^{[2]}) (I_{2m} - v M^{-1} \hat{F} \hat{\vartheta}) - \\
&\quad - (I_{2m} - v M^{-1} \hat{F} \hat{\vartheta}) (\lambda^2 \Lambda + \lambda P + W^{[2]}).
\end{aligned}$$

Moreover, we compute

$$\begin{aligned}
P' &= P + S_1^+ \Lambda - \Lambda S_1^+ = P - v M^{-1} \hat{\vartheta} \Lambda + \Lambda v M^{-1} \hat{\vartheta}, \\
P'_x &= P_x + S_{1,x}^+ \Lambda - \Lambda S_{1,x}^+, \\
S_{1,x}^+ &= -v_x M^{-1} \hat{\vartheta} + v M^{-1} M_x M^{-1} \hat{\vartheta} - v M^{-1} \hat{\vartheta}_x \\
&= -(\Lambda v D + P v) M^{-1} \hat{\vartheta} + v M^{-1} \hat{\vartheta} \Lambda v M^{-1} \hat{\vartheta} + v M^{-1} (\hat{D} \hat{\vartheta} \Lambda + \hat{\vartheta} P), \\
W^{[2]'} &= -\frac{1}{2} \Lambda (P'^2 - P'_x).
\end{aligned}$$

Substituting these expressions, $W^{[2]}$ from (56) and $D, \hat{D}$ from (54), into the formulas for S_t^+ and $V' S^+ - S^+ V$, rewriting them in terms of $v, \hat{v}, F, \hat{F}, M, P$, and P_x , and applying the identity (47), one can verify, through a similar, though more involved calculation, that

the temporal condition (36) holds. Therefore, the temporal condition of the binary DT is satisfied, completing the proof. $\square$

3.2. The Reduced Case

The transformed potential P' defined by (35) must satisfy the symplectic symmetry condition associated with $\text{sp}(2m)$:

$$\Delta P' \Delta^{-1} = -(P')^T, \quad (57)$$

where Δ is defined in (16). Equivalently, in terms of the binary DT in (49) and using (42), this reduction condition can be expressed as

$$\Delta[S_1^+, \Lambda] \Delta^{-1} = [(S_1^+)^T, \Lambda], \quad (58)$$

where S_1^+ is given explicitly by (48).

To the end, we choose the adjoint eigenvalues

$$\hat{\lambda}_i = \lambda_i, \quad 1 \leq i \leq N, \quad (59)$$

and define the adjoint eigenfunctions according to the symmetry of U and V :

$$\hat{v}_i = v_i^T \Delta^T = -v_i^T \Delta, \quad 1 \leq i \leq N. \quad (60)$$

We impose the orthogonality condition

$$\hat{v}_j v_i = 0, \quad \text{when } \hat{\lambda}_j = \lambda_i, \quad 1 \leq i, j \leq N, \quad (61)$$

to guarantee that the two Darboux matrices S^+ and S^- are inverses of each other. For $\hat{\lambda}_j \neq \lambda_i$, one can verify that

$$m_{ij} = \frac{\hat{v}_j v_i}{\lambda_j - \lambda_i} = -\frac{v_j^T \Delta v_i}{\lambda_j - \lambda_i} = -\frac{v_i^T \Delta v_j}{\lambda_i - \lambda_j} = \frac{\hat{v}_i v_j}{\lambda_i - \lambda_j} = m_{ji}.$$

For $\hat{\lambda}_j = \lambda_i$, $1 \leq i, j \leq N$, we have

$$(m_{ij}^c - m_{ji}^c)_x = (m_{ij}^c - m_{ji}^c)_t = 0, \quad (62)$$

so that we can demand $M^T = M$. Then, from (48), it follows that

$$\Delta S_1^+ \Delta^{-1} = -(S_1^+)^T, \quad (63)$$

and together with

$$\Delta \Lambda \Delta^{-1} = -\Lambda, \quad (64)$$

this guarantees (58) and hence the reduction property (57) for the transformed potential matrix P' .

We summarize the result in the following theorem.

Theorem 2. Let the adjoint eigenvalues $\{\hat{\lambda}_i \mid 1 \leq i \leq N\}$ and the adjoint eigenfunctions $\{\hat{v}_i \mid 1 \leq i \leq N\}$ be defined by (59) and (60), respectively. Assume that M is defined by (27), and that whenever $\hat{\lambda}_j = \lambda_i$ for $1 \leq i, j \leq N$, the elements m_{ij}^c of M are determined by (28) and (29), satisfying

$$m_{ij}^c = m_{ji}^c, \quad \text{when } \hat{\lambda}_j = \lambda_i, \quad 1 \leq i, j \leq N, \quad (65)$$

so that $M^T = M$. If, in addition, the orthogonality conditions (61) are satisfied, then (49) defines a binary DT for the symmetric matrix integrable NLS model (12).

3.3. A Demonstrative Example

Let us now consider a case $m = N = 2$ and present an instructive example. We take the zero-seed solution, so that the associated eigenfunctions are defined by

$$v_i = (e^{\lambda_i x + \lambda_i^2 t} \xi_1^{[i]T}, e^{-\lambda_i x - \lambda_i^2 t} \xi_2^{[i]T})^T, \quad 1 \leq i \leq 2, \quad (66)$$

and the adjoint eigenfunctions are determined by

$$\hat{v}_i = -v_i^T \Delta, \quad 1 \leq i \leq 2, \quad (67)$$

where $\xi_j^{[i]}, 1 \leq i, j \leq 2$, are arbitrary complex two-dimensional column vectors.

Since we take

$$\hat{\lambda}_i = \lambda_i, \quad 1 \leq i \leq 2, \quad (68)$$

it is required to impose

$$\hat{v}_i v_i = 0, \quad 1 \leq i \leq 2, \quad (69)$$

which holds automatically. Furthermore, if we take

$$m_{11}^c = m_{22}^c = 0, \quad (70)$$

two additional sets of orthogonality conditions, arising from the definitions (28)–(30) of the entries m_{ij}^c , must be taken into account. The first set is

$$\hat{v}_i \Delta v_i = 0, \quad 1 \leq i \leq 2. \quad (71)$$

Equivalently, this yields

$$\xi_1^{[i]T} \xi_2^{[i]} = 0, \quad 1 \leq i \leq 2. \quad (72)$$

The second set is

$$2\lambda_i \hat{v}_i \Delta v_i = 0, \quad 1 \leq i \leq 2, \quad (73)$$

which follows directly from (71). Therefore, it suffices to impose only the conditions in (72).

Observing

$$p' = [S_1^+, \Lambda]_{12} = -2(S_1^+)_{12}, \quad q' = [S_1^+, \Lambda]_{21} = 2(S_1^+)_{21},$$

we can explicitly determine

$$\begin{cases} p'_{1,1} = 4(\lambda_1 - \lambda_2) \frac{\xi_{1,1,1} \xi_{2,1,1}}{h_1}, & p'_{2,2} = 4(\lambda_1 - \lambda_2) \frac{\xi_{1,1,2} \xi_{2,1,2}}{h_1}, \\ p'_{1,2} = p'_{2,1} = 2(\lambda_1 - \lambda_2) \frac{\xi_{1,1,1} \xi_{2,1,2} + \xi_{1,1,2} \xi_{2,1,1}}{h_1}, \end{cases} \quad (74)$$

and

$$\begin{cases} q'_{1,1} = 4(\lambda_1 - \lambda_2) \frac{\xi_{1,2,1} \xi_{2,2,1}}{h_2}, & q'_{2,2} = 4(\lambda_1 - \lambda_2) \frac{\xi_{1,2,2} \xi_{2,2,2}}{h_2}, \\ q'_{1,2} = q'_{2,1} = 2(\lambda_1 - \lambda_2) \frac{\xi_{1,2,1} \xi_{2,2,2} + \xi_{1,2,2} \xi_{2,2,1}}{h_2}, \end{cases} \quad (75)$$

where

$$\begin{cases} h_1 = (\xi_{1,2,1}\xi_{2,1,1} + \xi_{1,2,2}\xi_{2,1,2})e^{-2(\lambda_1 x + \lambda_1^2 t)} - (\xi_{1,1,1}\xi_{2,2,1} + \xi_{1,1,2}\xi_{2,2,2})e^{-2(\lambda_2 x + \lambda_2^2 t)}, \\ h_2 = (\xi_{1,2,1}\xi_{2,1,1} + \xi_{1,2,2}\xi_{2,1,2})e^{2(\lambda_2 x + \lambda_2^2 t)} - (\xi_{1,1,1}\xi_{2,2,1} + \xi_{1,1,2}\xi_{2,2,2})e^{2(\lambda_1 x + \lambda_1^2 t)}. \end{cases}$$

In the above formulas, we denote

$$p' = (p'_{ij})_{2 \times 2}, q' = (q'_{ij})_{2 \times 2}, \xi_j^{[i]} = (\xi_{i,j,1}, \xi_{i,j,2})^T, 1 \leq i, j \leq 2. \quad (76)$$

With the orthogonality conditions in (72), these functions explicitly solve the matrix integrable NLS model with symmetric matrix potentials in (12).

4. Concluding Remarks

This study investigates binary DTs for the integrable NLS system with two symmetric matrix potentials via the M -matrix generating scheme. These DT structures play a fundamental role in understanding the integrable dynamics of the matrix NLS model. Starting from the vacuum state, successive applications of the resulting binary DTs yield a wide variety of explicit soliton solutions.

The concrete examples presented here provide coupled systems of nonlinear integrable equations, demonstrating the practical applicability of the theoretical framework and highlighting the rich integrability structures of these systems. In particular, the M -matrix must be extended to accommodate cases where eigenvalues coincide with adjoint eigenvalues, rendering the classical formulation of the M -matrix inapplicable.

Based on the eigenvalues of the associated matrix eigenvalue problems, soliton solutions can be classified into different types, such as negatons, positons, and complexitons. These solutions can be further categorized into solitons, breathers, kinks, anti-kinks, lump waves, rogue waves, and various mixed interaction solutions. Notably, interaction solutions may involve arbitrary functions. Some of these solutions arise through specific reductions of wave numbers in general soliton solutions (see, e.g., [24,25]). Moreover, lump wave solutions have recently been extensively studied for nonlinear models in (2+1)-dimensions (see, e.g., [26–29]). In particular, interaction solutions may have potential applications in describing rapid motions, such as jumping, which involve the rapid conversion of potential energy into kinetic energy, as observed in robotic systems [30].

As the number of dependent variables increases, or as higher powers of the spectral parameter appear in the spatial spectral matrix, progressively larger and more intricate integrable systems can be constructed (see, e.g., [31,32]). The resulting complexity, particularly with the emergence of nonlocalities, poses additional analytical challenges. Nevertheless, the classification of these extended integrable systems and the study of their soliton structures continue to provide valuable insights into the fundamental mechanisms governing nonlinear dynamics and integrability in mathematical physics [33].

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